

Showcasing Predictions on the User Interface

The Flask web application provides an interactive interface that enables users to predict the Human Development Index (HDI) using the trained machine learning model. The application consists of three main pages: the Home Page, the Prediction Page, and the Result Page.

Home Page (home.html):

The home page introduces the Human Development Index (HDI), explains its importance, and provides an overview of the prediction system. A **Predict** button in the navigation bar redirects the user to the prediction page.

Prediction Page (indexnew.html):

The prediction page contains a form where users can select a country from a dropdown list and enter feature values such as Life Expectancy, Mean Years of Schooling, Gross National Income (GNI) per Capita, and Internet Users. After filling in the required details, clicking the **Predict** button submits the form to the Flask application.

Result Page (resultnew.html):

The trained Linear Regression model processes the submitted values and predicts the HDI score. The predicted value is displayed along with its corresponding category (Low HDI, Medium HDI, High HDI, or Very High HDI), allowing users to easily interpret the result.

```
49 </style>
50 </head>
51 <body>
52 <div class="navbar">
53 <a href="/Prediction" >Predict</a>
54 <a href="/Home">Home</a>
55 <br>
56 </div>
57 <br>
58 <center><b><font color="white" size="15" font-family="Comic Sans MS" >Human Development Index</font></b></center>
59 <div>
60 <br>
61 <center>
62 <p>The Human Development Index (HDI) is a statistic composite index of life expectancy, education,
63 and per capita income indicators, which are used to rank countries into four tiers (very high, high, medium & low)
64 of human development. A country scores a higher HDI when the lifespan is higher, the education level is higher,
65 and the gross national income GNI (PPP) per capita is higher. The HDI was created to emphasize that people and their
66 capabilities should be the ultimate criteria for assessing the development of a country, not economic growth alone.
67 The HDI can also be used to question national policy choices, asking how two countries with the same level of
68 GNI per capita can end up with different human development outcomes. These contrasts can stimulate debate about
69 government policy priorities. In order to do this we will be building a machine learning model to predict the
70 Human Development Index of a country by taking few important aspects as inputs. Our model will at last predict
71 the HDI score of a country and will also tell under which category will it be falling (very high, high, medium or
72
73 </p>
74 </center>
75 </div>
76 </body>
77 </html>
```

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108 <body>
109 <center><div class="header1" <font color="red" font-family="Fascinate Inline" size=7></font>human development
110
111
112 <br><br><br><br><br>
113 <form class="main" action="/predict" method="post">
114 <br>
115 <center><select id="Country" name="Country">
116 <option value="">Select the name of the Country</option>
117 <option value="0">Afghanistan</option>
118 <option value="8">Australia</option>
119 <option value="13">Bangladesh</option>
120 <option value="31">Canada</option>
121 <option value="76">India</option>
122 <option value="138">Poland</option>
123 <option value="179">Turkey</option>
124 </select></center>
125
126
127 <input class="form-input" type="text" name='Life expectancy' placeholder="Enter life expectancy rate(range 50-89)">
128
129 <input class="form-input" type="text" name='Mean years of schooling' placeholder="Enter Mean years of schooling(rang
130
131
132 <input class="form-input" type="text" name='Gross national income (GNI) per capita' placeholder="Enter GNI(range 290
133
134
135 <input class="form-input" type="text" name='Internet users' placeholder="Enter the number of internet users"><br>
136 <center><input type="submit" class="logbtn" value="Predict"></center>
137 <div class="bor"><center><b><font color="white" size=5>{{showcase}}</font></b></center></div>
138 </form>
139
140
141 </body>

```

SHOWCASING PREDICTIONS ON THE UI

The web application consists of three pages: **Home Page**, **Prediction Page**, and **Result Page**. The application is built using Flask and HTML templates.

1. Home Page (home.html)

The home page provides a brief introduction to the Human Development Index (HDI) and its importance. A "Predict" button is available in the top-right corner which navigates the user to the Prediction page.



2. Prediction Page (indexnew.html)

The prediction page allows users to input values for the required features. Users can select a country from the dropdown and enter the following details within the valid range. After entering all the values, clicking the 'Predict' button will submit the form and process the input.

The screenshot shows the Prediction Page of the Human Development Index application. It features a dark blue header with the title "Human Development Index" and a "Predict" button in the top right corner. The main content area has a dark blue background with white text. It contains a form with the following fields: Country (dropdown menu), Life expectancy (range 50-89), Mean years of schooling (range 1-15), Gross national income (GNI) per capita (range 290-85388), and Internet users (range 0-100000). A "Predict" button is located at the bottom right of the form.

3. Result Page (resultnew.html)

After clicking the 'Predict' button, the result page displays the predicted HDI score along with the category under which the country falls (Low, Medium, High, or Very High HDI).



4. Web Page Overview

